New supercomputer is a rack of PlayStations

When the PlayStation3 was released in November 2006, Gaurav Khanna's wife braved long queues so he could be one of the first people in the US to get his hands on the gaming console.

But the astrophysicist was not itching to burn some rubber in *Gran Turismo* or shoot hoops in *NBA 07*. Instead he wanted to build his own supercomputer.

Mr Khanna now owns 16 PS3s, which spend their days simulating the activities of very large black holes in the universe for the physics department at the University of Massachusetts.

Hooked together in a single cluster, the PS3 consoles provide his department with the same amount of computing power as a 400-node supercomputer.

"The challenge these days with supercomputing facilities is that there is a lot of demand for them. So even if I submitted a job that would be expected to take about an hour, it could actually take two days to get started because the queues are so long.

"The PS3 cluster is all mine and was very low cost to set up, which makes it really attractive," he says.

What makes the gaming console vastly superior to high-end computers for complex research algorithms, Mr Khanna says, is the Cell chip built by IBM to facilitate high-end gaming functions on the latest generation of consoles.

In addition the PS3 was built with an open hardware architecture, which can run the Linux operating system.

Based on Einstein's theory of relativity, Mr Khanna's research on black holes is purely theoretical. In order to run his simulation data on the console he has to reprogram it so the algorithms will work on the new architecture.

"Linux can turn any system into a general purpose computer but for it to do work for me I have to run my own code on it for astrophysics applications. The hard part of the job was to make sure my own calculations could run fast on the platform, which meant I had to optimise the written code so it could utilise the new features of the system.

"I am not a Linux person - I am a Mac person - but I was able to follow instructions online," he says.

His next challenge will be to turn his data into graphical simulations using the high end graphics engine included in the PS3.

"We haven't done that yet but it would be very neat to actually see the simulation while it is going on," he says.

Although Mr Khanna was one of the first scientists to optimise the PS3 for his own research work, Tod Martinez, Professor of Chemistry at the University of Illinois has been tinkering with games consoles ever since his son's original PlayStation malfunctioned.